



Line Pressure Surveillance

August 7th, 2018

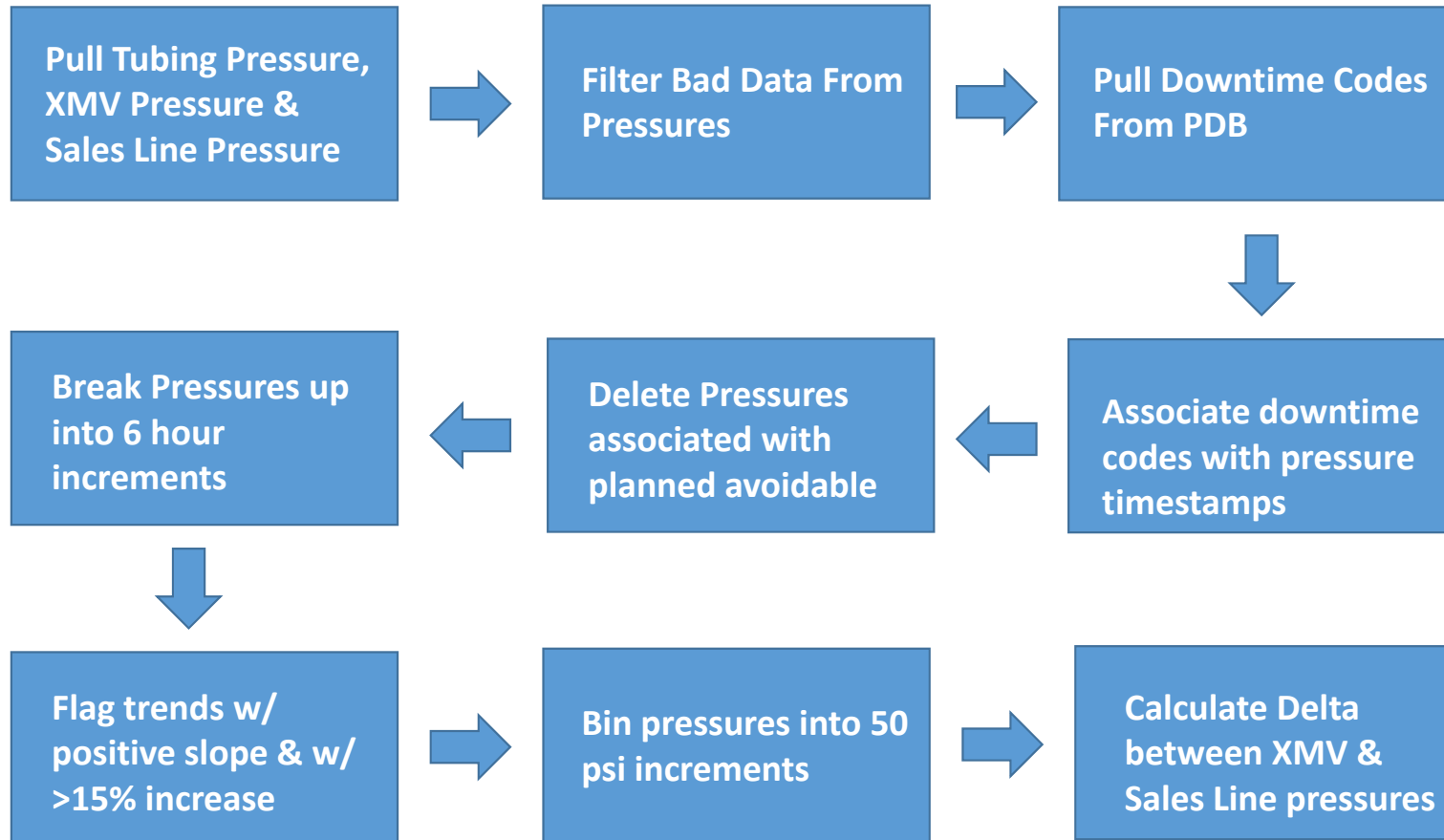


Agenda

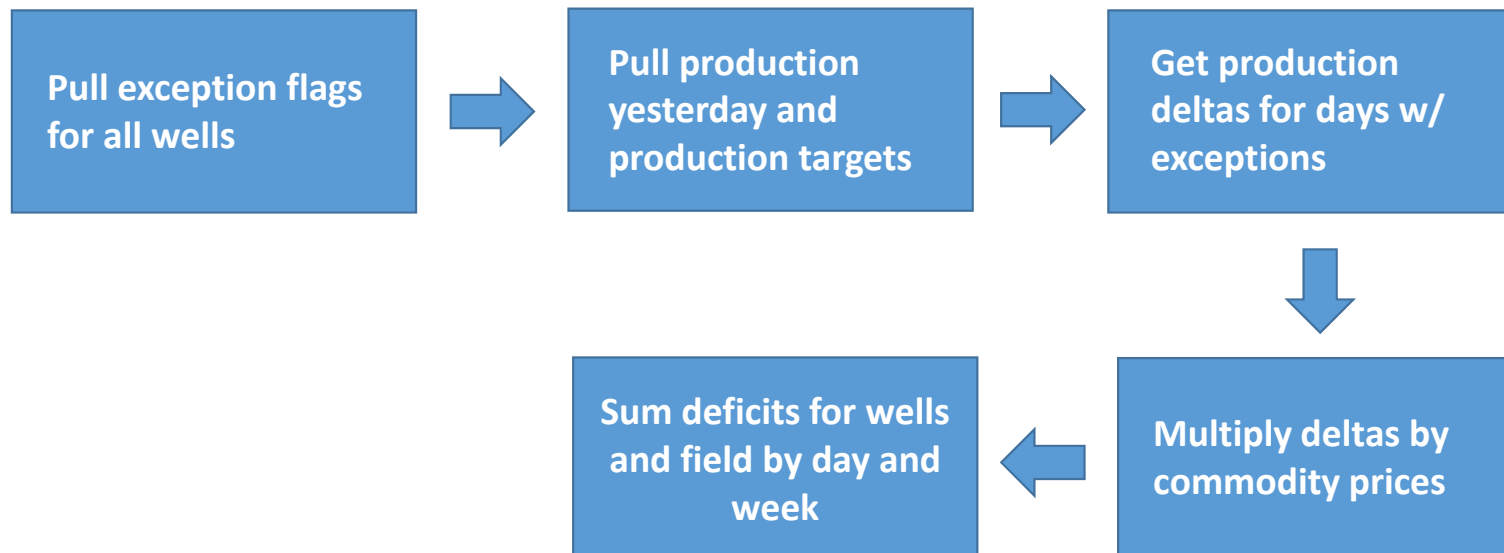


- Workflow Steps
- Workflow Outputs
- Physics Based Modeling Review
 - IPR/VLP & Mach Number
- JMZ 1C-3HZ Model
 - Early Life & Today
- Path Forward

Workflow 1 Steps



Workflow 2 Steps



XMV Pressure Exception



Field Oil Differed (BBLs): 12,832

Field Sales Differed (\$): \$898,274

XMV Pressure Exception

Gas Sales Exception XMV to Line Delta Exception Summary

Well Name	Tubing Pressure Bin	XMV Pressure Bin	Gas Sales Pressure Bin	XMV to Sales Press Delta	8/1/2018 4:00:00 AM	8/2/2018 4:00:00 AM	8/3/2018 4:00:00 AM	8/4/2018 4:00:00 AM	8/5/2018 4:00:00 AM	8/6/2018 4:00:00 AM	8/7/2018 4:00:00 AM	Totals
Wilson Ranch 33C-27HZ	500 psi - 550 psi	150 psi - 200 psi	100 psi - 150 psi	45	0	101	0	0	0	0	0	101
					\$0	\$7,044	\$0	\$0	\$0	\$0	\$0	\$7,044
Gittlein 3C-28HZ	300 psi - 350 psi	100 psi - 150 psi	100 psi - 150 psi	22	13	14	13	14	14	10	21	99
					\$937	\$950	\$889	\$948	\$985	\$728	\$1,498	\$6,935
Jaguar 3C-14HZ	850 psi - 900 psi	100 psi - 150 psi	150 psi - 200 psi	-42	0	0	0	0	0	0	88	88
					\$0	\$0	\$0	\$0	\$0	\$0	\$6,160	\$6,160
Weichel 30N-23HZ	150 psi - 200 psi	150 psi - 200 psi	150 psi - 200 psi	0	0	15	9	31	17	8	8	88
					\$0	\$1,033	\$602	\$2,181	\$1,179	\$539	\$594	\$6,128
Bbkb 16N-A35HZ	900 psi - 950 psi	150 psi - 200 psi	150 psi - 200 psi	1	0	0	0	30	9	0	48	87
					\$0	\$0	\$0	\$2,133	\$653	\$0	\$3,327	\$6,113
Ss Minnow 23N-14HZ	950 psi - 1000 psi	150 psi - 200 psi	150 psi - 200 psi	1	48	0	0	0	0	0	39	86
					\$3,335	\$0	\$0	\$0	\$0	\$0	\$2,720	\$6,055
Small Eyed 14C-35HZ	750 psi - 800 psi	100 psi - 150 psi	100 psi - 150 psi	25	0	0	0	84	0	0	0	84
					\$0	\$0	\$0	\$5,904	\$0	\$0	\$0	\$5,904
Wardell 28N-20HZ	100 psi - 150 psi	150 psi - 200 psi	150 psi - 200 psi	11	10	12	12	15	8	12	11	80
					\$671	\$840	\$869	\$1,017	\$586	\$847	\$797	\$5,626
Underhill State 26C-17HZ	150 psi - 200 psi	100 psi - 150 psi	100 psi - 150 psi	9	0	3	17	20	20	0	20	80
					\$0	\$203	\$1,211	\$1,396	\$1,400	\$0	\$1,378	\$5,587
Dove 8C-13HZ	100 psi - 150 psi	100 psi - 150 psi	100 psi - 150 psi	10	5	0	20	21	0	27	0	72
					\$329	\$0	\$1,376	\$1,436	\$0	\$1,909	\$0	\$5,051
Paragon 40C-31HZ	100 psi - 150 psi	100 psi - 150 psi	100 psi - 150 psi	10	0	24	24	0	0	0	24	72
					\$0	\$1,669	\$1,683	\$0	\$0	\$0	\$1,672	\$5,023
Melbon 19C-8HZ	100 psi - 150 psi	100 psi - 150 psi	0 psi - 100 psi	15	17	15	12	8	8	0	12	71
					\$1,156	\$1,079	\$838	\$535	\$531	\$0	\$863	\$5,002
Highway 160 37N-2HZ	100 psi - 150 psi	100 psi - 150 psi	100 psi - 150 psi	4	0	0	4	27	0	22	18	71
					\$0	\$0	\$249	\$1,912	\$0	\$1,527	\$1,266	\$4,954
Troutt 14N-33Hz	200 psi - 250 psi	150 psi - 200 psi	100 psi - 150 psi	43	12	6	2	15	19	0	17	70
					\$826	\$393	\$145	\$1,038	\$1,309	\$0	\$1,208	\$4,918
Cream State 28N-21HZ	150 psi - 200 psi	100 psi - 150 psi	100 psi - 150 psi	40	15	13	12	5	3	9	12	70
					\$1,070	\$930	\$861	\$352	\$226	\$641	\$820	\$4,901

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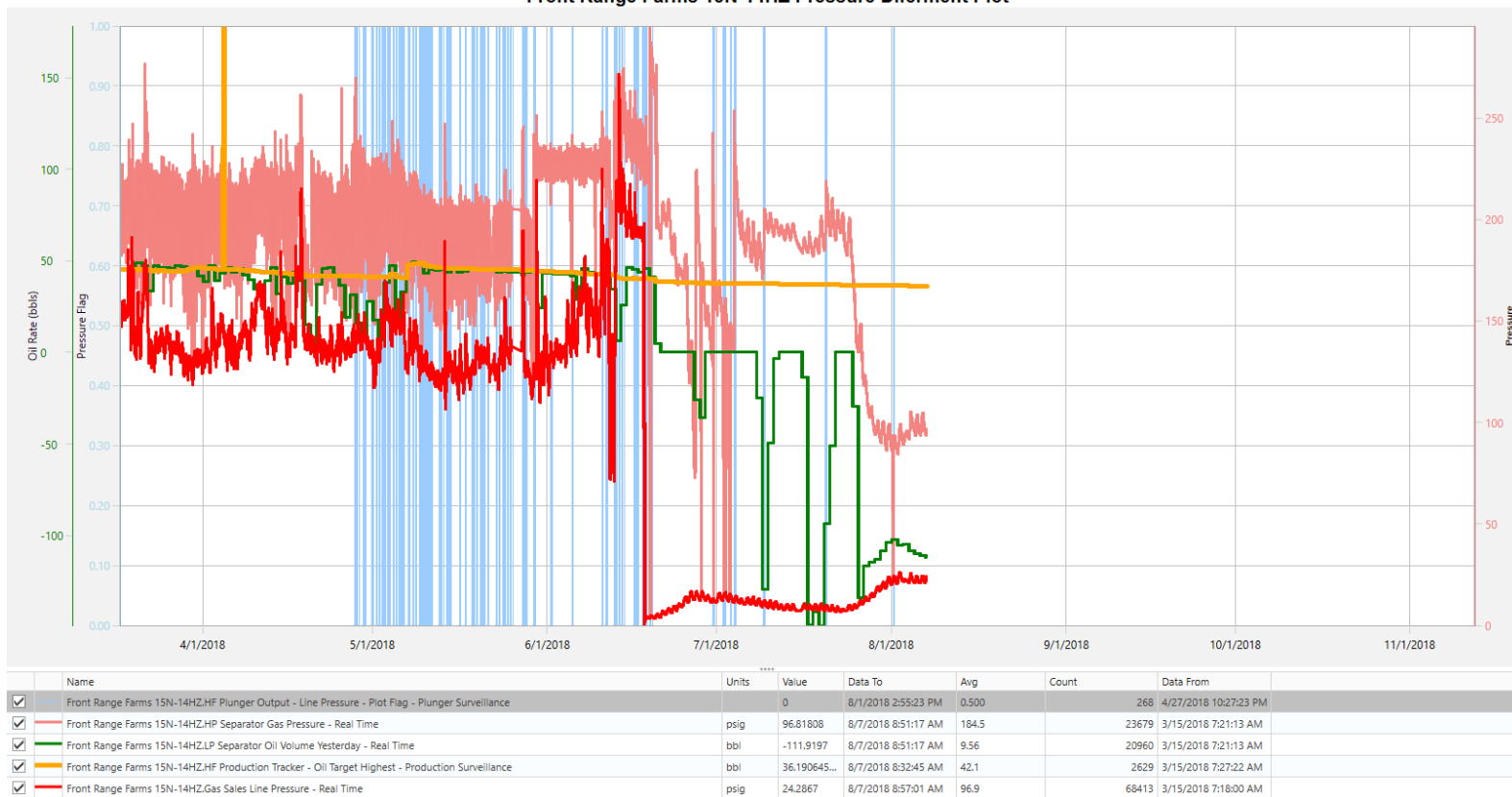
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XMV Pressure Exception



Front Range Farms 15N-14HZ Pressure Diferment Plot



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Gas Sales Pressure Exception



Field Oil Differed (BBLs): 14,763
Field Sales Differed (\$): \$1,033,382

Gas Sales Pressure Exception

XMV Pressure Exception XMV to Line Delta Exception Summary

Well Name	Tubing Pressure Bin	XMV Pressure Bin	Gas Sales Pressure Bin	XMV to Sales Press Delta	8/1/2018 4:00:00 AM	8/2/2018 4:00:00 AM	8/3/2018 4:00:00 AM	8/4/2018 4:00:00 AM	8/5/2018 4:00:00 AM	8/6/2018 4:00:00 AM	8/7/2018 4:00:00 AM	Totals
Whisper Rock 3C-25HZ	950 psi - 1000 psi	150 psi - 200 psi	100 psi - 150 psi	62	0	0	77	0	0	67	0	144
					\$0	\$0	\$5,378	\$0	\$0	\$4,667	\$0	\$10,046
Burrough 17N-17HZ	900 psi - 950 psi	150 psi - 200 psi	150 psi - 200 psi	-1	0	79	0	0	0	0	0	79
					\$0	\$5,534	\$0	\$0	\$0	\$0	\$0	\$5,534
Cream 15N-A28HZ	550 psi - 600 psi	0 psi - 100 psi	100 psi - 150 psi	-34	28	28	27	28	27	0	27	137
					\$1,940	\$1,928	\$1,925	\$0	\$1,913	\$0	\$1,875	\$9,581
Cannon State 34C-27HZ	200 psi - 250 psi	100 psi - 150 psi	100 psi - 150 psi	15	0	40	35	0	0	0	36	111
					\$0	\$2,798	\$2,452	\$0	\$0	\$0	\$2,554	\$7,804
Weichel 26N-11HZ	400 psi - 450 psi	150 psi - 200 psi	150 psi - 200 psi	0	27	26	26	27	27	27	28	186
					\$1,855	\$1,819	\$1,849	\$1,740	\$1,907	\$1,965	\$1,891	\$13,026
Weichel 1N-11HZ	300 psi - 350 psi	150 psi - 200 psi	150 psi - 200 psi	1	25	23	25	25	22	23	23	160
					\$1,719	\$1,629	\$1,749	\$1,334	\$1,554	\$1,604	\$1,592	\$11,182
Garfield 15C-35HZ	550 psi - 600 psi	200 psi - 250 psi	150 psi - 200 psi	65	0	0	30	0	0	0	0	30
					\$0	\$0	\$2,099	\$0	\$0	\$0	\$0	\$2,099
Camenisch 29C-35HZ	600 psi - 650 psi	200 psi - 250 psi	150 psi - 200 psi	65	0	0	0	0	0	0	50	50
					\$0	\$0	\$0	\$0	\$0	\$0	\$3,507	\$3,507
Bbkb 15N-35HZ	950 psi - 1000 psi	150 psi - 200 psi	150 psi - 200 psi	0	48	27	26	48	23	0	64	189
					\$3,389	\$1,895	\$1,848	\$0	\$1,639	\$0	\$4,452	\$13,223
Wagner 13C-3HZ	150 psi - 200 psi	150 psi - 200 psi	150 psi - 200 psi	15	0	0	0	0	0	0	18	18
					\$0	\$0	\$0	\$0	\$0	\$0	\$1,253	\$1,253
Boyd 1N-32HZ	100 psi - 150 psi	100 psi - 150 psi	100 psi - 150 psi	16	0	20	21	0	0	0	22	63
					\$0	\$1,397	\$1,468	\$0	\$0	\$0	\$1,562	\$4,427
Highlands 28N-33HZ	250 psi - 300 psi	200 psi - 250 psi	200 psi - 250 psi	41	0	0	116	0	0	0	115	231
					\$0	\$0	\$8,138	\$0	\$0	\$0	\$8,053	\$16,191
Bbkb 37N-35HZ	850 psi - 900 psi	150 psi - 200 psi	150 psi - 200 psi	1	0	0	42	0	53	0	76	172
					\$0	\$0	\$2,973	\$0	\$3,736	\$0	\$5,324	\$12,033
Wilson Ranch 33C-27HZ	500 psi - 550 psi	150 psi - 200 psi	100 psi - 150 psi	45	0	101	30	0	0	0	0	131
					\$0	\$7,044	\$2,127	\$0	\$0	\$0	\$0	\$9,171
Gittlein 3C-28HZ	300 psi - 350 psi	100 psi - 150 psi	100 psi - 150 psi	22	0	14	13	0	0	0	21	48
					\$0	\$950	\$889	\$0	\$0	\$0	\$1,498	\$3,337

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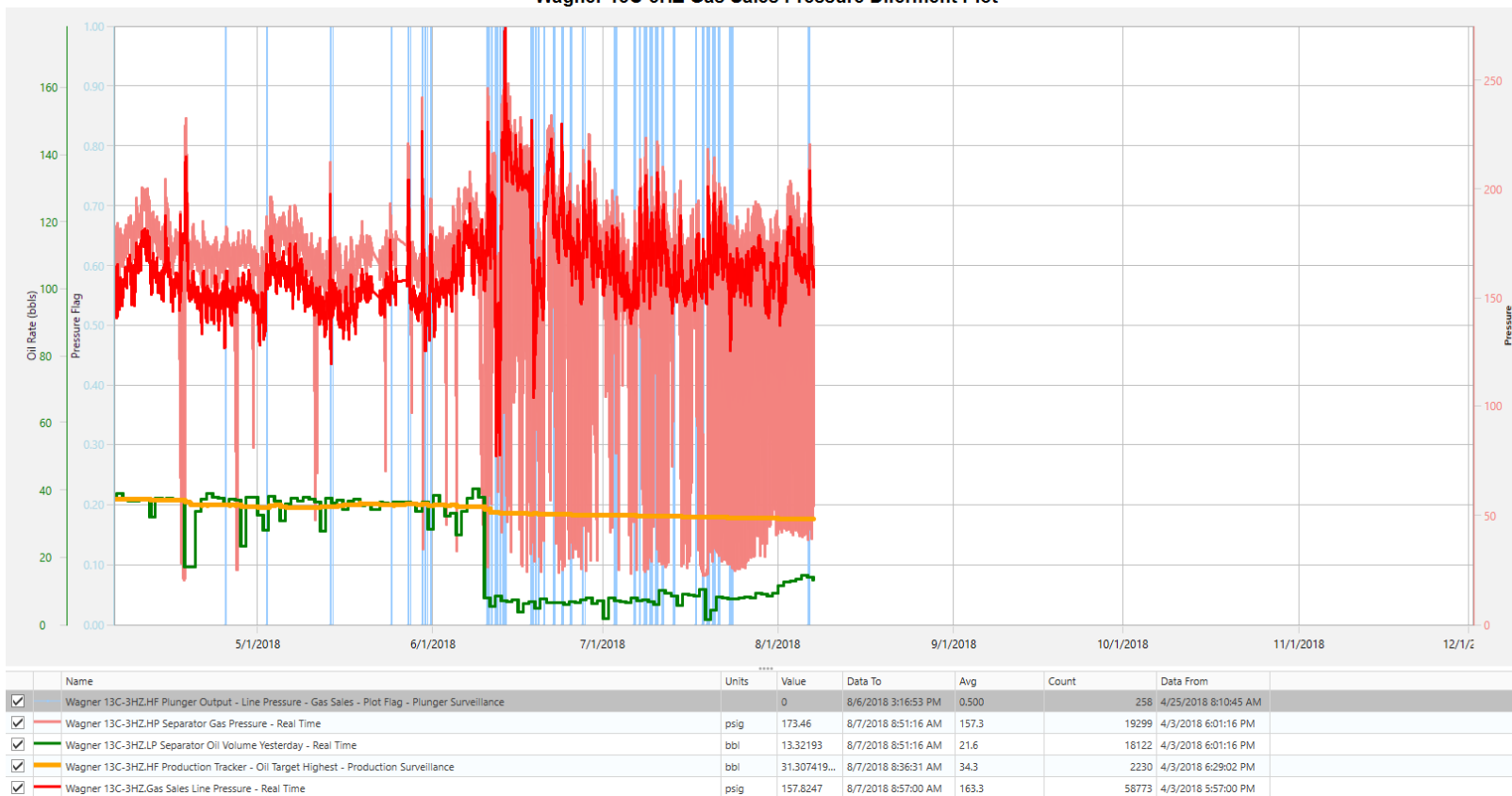
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Gas Sales Pressure Exception



Wagner 13C-3HZ Gas Sales Pressure Diferment Plot



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XMV to Gas Sales Line Delta



Field Oil Differed (BBLS):

12,832

Field Sales Differed (\$):

\$898,274

XMV to Gas Sales Line Delta Exception

Gas Sales Exception

Line Pressure Summary

Exception Summary

Well Name	Tubing Pressure Bin	XMV Pressure Bin	Gas Sales Pressure Bin	XMV to Sales Press Delta	8/1/2018 4:00:00 AM	8/2/2018 4:00:00 AM	8/3/2018 4:00:00 AM	8/4/2018 4:00:00 AM	8/5/2018 4:00:00 AM	8/6/2018 4:00:00 AM	8/7/2018 4:00:00 AM	Totals
Powers 1C-23HZ	0 psi - 100 psi	100 psi - 150 psi	0 psi - 100 psi	126	0	0	0	0	0	0	0	0
					\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Swartz 1N-12HZ	950 psi - 1000 psi	200 psi - 250 psi	100 psi - 150 psi	95	0	0	0	0	0	0	0	0
					\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Swartz 26C-12HZ	600 psi - 650 psi	200 psi - 250 psi	100 psi - 150 psi	95	0	0	0	0	0	0	0	0
					\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Swartz 26N-25HZ	500 psi - 550 psi	200 psi - 250 psi	100 psi - 150 psi	94	0	0	0	0	0	0	0	0
					\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Swartz 1C-25HZ	550 psi - 600 psi	200 psi - 250 psi	100 psi - 150 psi	93	0	0	0	0	0	0	0	0
					\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Swartz 27N-25HZ	500 psi - 550 psi	200 psi - 250 psi	100 psi - 150 psi	89	0	0	0	0	0	0	0	0
					\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Swartz 27C-12HZ	600 psi - 650 psi	200 psi - 250 psi	100 psi - 150 psi	89	0	0	0	0	0	0	0	0
					\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
GEE 26C-12HZ	350 psi - 400 psi	200 psi - 250 psi	100 psi - 150 psi	87	0	0	0	0	0	0	0	0
					\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
GEE 16C-25HZ	300 psi - 350 psi	200 psi - 250 psi	100 psi - 150 psi	87	0	0	0	0	0	0	0	0
					\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
GEE 2C-12HZ	250 psi - 300 psi	200 psi - 250 psi	100 psi - 150 psi	86	0	0	0	0	0	0	0	0
					\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
GEE 15C-25HZ	300 psi - 350 psi	200 psi - 250 psi	100 psi - 150 psi	86	0	0	0	0	0	0	0	0
					\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
GEE 15N-25HZ	250 psi - 300 psi	200 psi - 250 psi	100 psi - 150 psi	85	0	0	0	0	0	0	0	0
					\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
GEE 27N-12HZ	300 psi - 350 psi	200 psi - 250 psi	100 psi - 150 psi	85	0	0	0	0	0	0	0	0
					\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
GEE 16N-25HZ	250 psi - 300 psi	200 psi - 250 psi	100 psi - 150 psi	85	0	0	0	0	0	0	0	0
					\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
GEE 37N-25HZXX	200 psi - 250 psi	200 psi - 250 psi	100 psi - 150 psi	84	0	0	0	0	0	0	0	0
					\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0

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XMV to Gas Sales Line Delta



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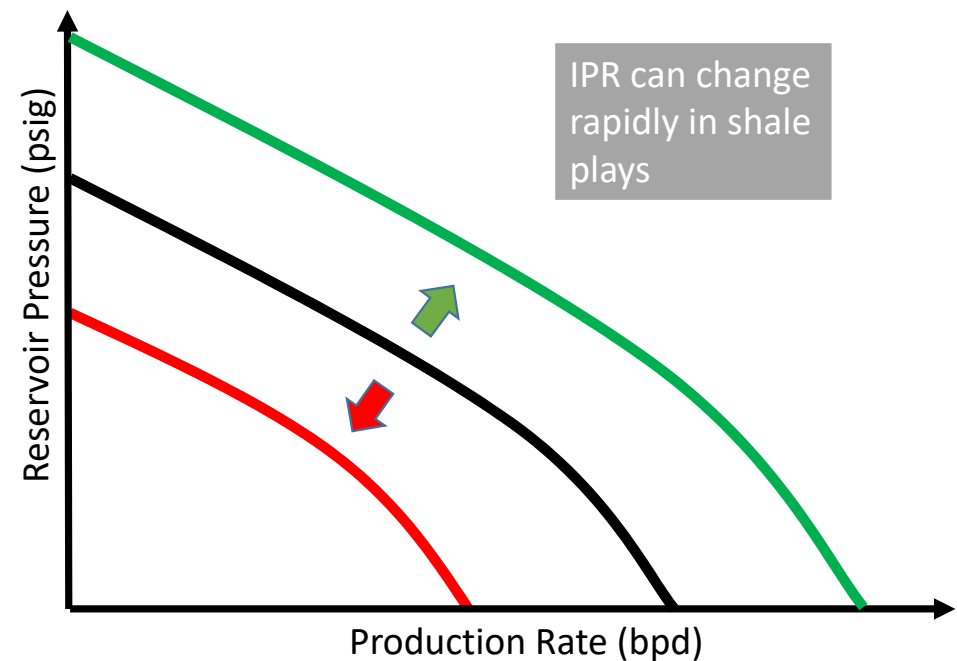
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Inflow Performance Relationship



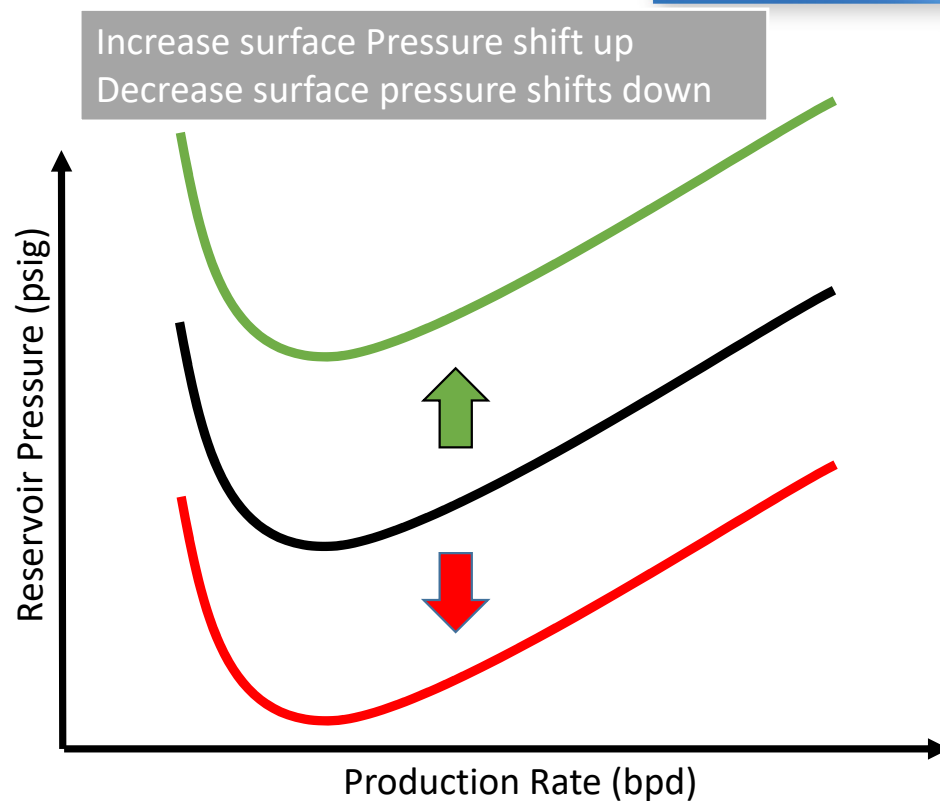
- Relationship Between:
 - Reservoir Pressure
 - Flowing Bottom Hole Pressure
 - Production Rate
- Stable in conventional plays
 - Sand Stone, etc.
- Dynamic in unconventional plays
 - Shale, etc.



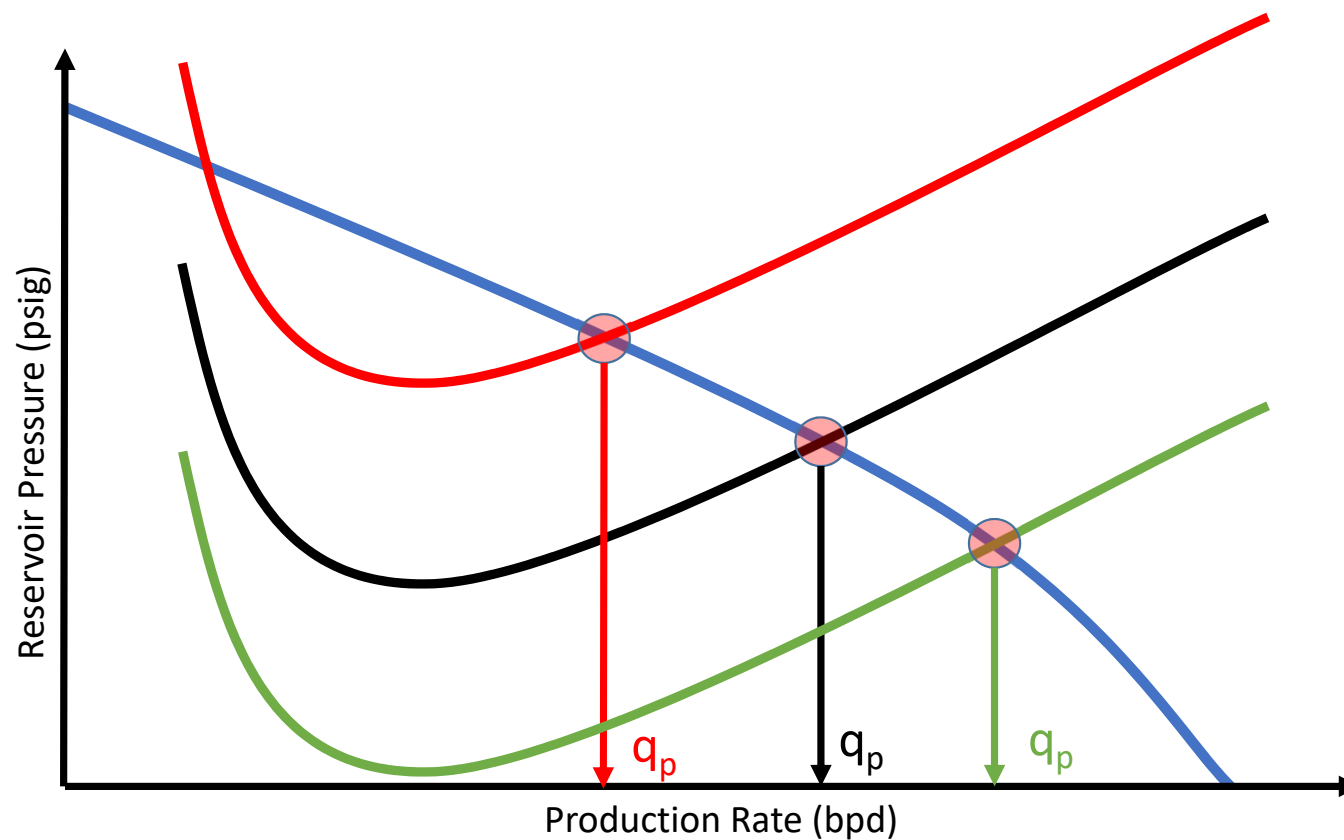
Vertical Lift Performance Relationship



- VLP: Bottom hole pressure as a function of flow rate.
- Dependent on:
 - Well Depth
 - Tubing Size
 - Water Cut
 - GOR
 - PVT Properties
 - **Surface Pressure**



Combining IPR & VLP



Intersection is operating point.

Changes in surface pressure change production rates with all other variables held constant

Compressible Flow

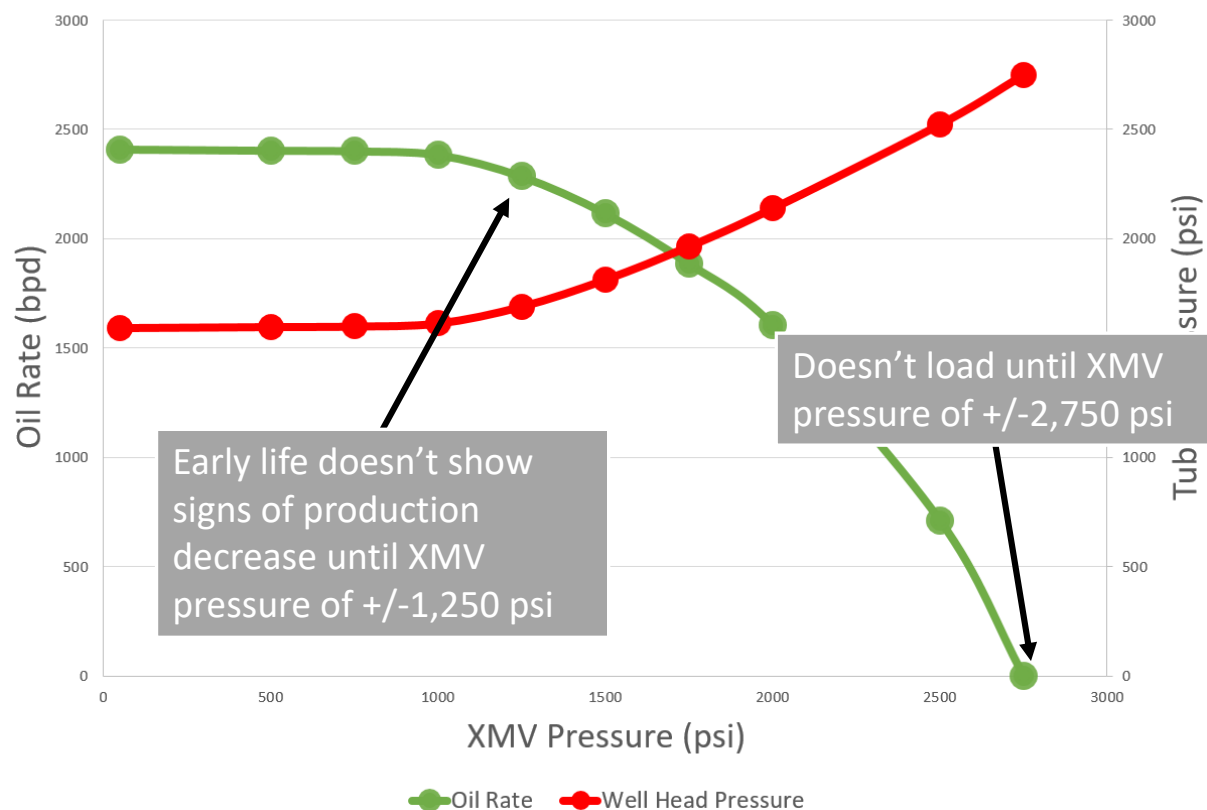


- Gas at subsonic flow acts like an incompressible fluid
 - If line pressure goes up, well head pressure goes up
- Gas at supersonic flow acts like a compressible fluid
 - If line pressure goes up, well head pressure isn't effected

XMV Pressure Sensitivity Example 1



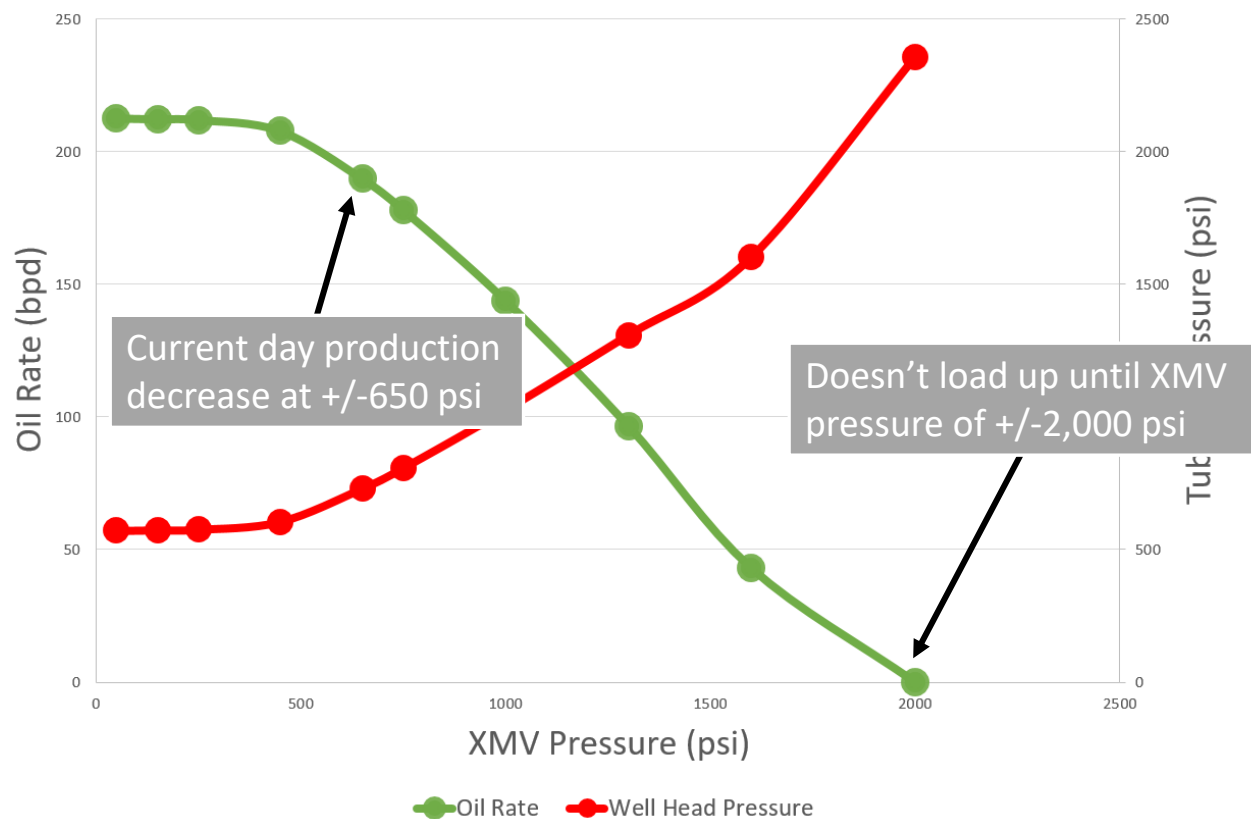
JZM 1C-3HZ XMV Pressure Sensitivity Plot



XMV Pressure Sensitivity Plot Example 2



JZM 1C-3HZ XMV Pressure Sensitivity Plot



Path Forward



- Too many dynamic variables for solution to be data driven
- Wells with high pressure immune to line pressure increases
- Solution requires physics based modeling
 - Models need to be ran once a week
- Physics based modeling can be automated in IVM
- Output data in SQL statement
 - Robert Palumbo currently working this
- Make edits to thresholds in workflow based off feedback
- Meet with Kreg about creating VLP's for all wells in the field
- Run macro model to see what lowering pressures do to production